

TMGA REGULATORY COMPLIANCE & MAPPING MATRIX

Cross-Standard Regulatory Compliance & Technical Mapping: Mapping TMGA's 33 Dimensions, NHCP, ZKP, Shadow Auditing & Circuit Breaker Protocols to NIST AI RMF 1.0 and ISO/IEC 42001:2023

Executive Summary & Regulatory Scope: The Mind Genesis Architecture (TMGA) bridges abstract alignment principles and concrete engineering runtime constraints. While conventional regulatory frameworks such as the **NIST AI Risk Management Framework (AI RMF 1.0)** and **ISO/IEC 42001:2023 (Artificial Intelligence Management System - AIMS)** define *what* operational controls and organizational processes must exist, TMGA supplies the *architectural mechanism* that implements these requirements at runtime. This document provides a formal, auditable compliance matrix mapping TMGA's 33 multidimensional vectors (AGI, ASI, AWI), Non-Hierarchical Consensus Protocol (NHCP), Zero-Knowledge Proof (ZKP) pipeline, Asynchronous Shadow Auditing, and Hardware-Assisted Epistemic Containment directly into global regulatory controls.

1. NIST AI RMF 1.0 Core Functions Mapping

NIST AI RMF 1.0 organizes AI risk management into four core functions: **GOVERN, MAP, MEASURE, and MANAGE**. The table below details how TMGA's runtime mechanisms satisfy each function's primary sub-categories.

NIST RMF Core	Sub-Category & Requirement	TMGA Architectural Mapping	Technical Enforcement Mechanism
GOVERN (Gov 1.1–6.2)	Policies, processes, and accountability structures for AI risk, compliance, and ethical oversight.	Sphere III: AWI AWI Systems (3.9) AWI Safe (3.6) Genesis Ethic	Continuous runtime governance through $CAWI(St+1) = TRUE$ validation, Non-Hierarchical Consensus Protocol (NHCP), and Cognitive Sovereignty rules.
MAP (Map 1.1–5.2)	Context identification, categorization of AI capabilities, task boundaries, and hazard mapping.	Sphere I & III AWI Mind (3.1) AGI/ASI Fold (1.4/2.4) AWI Void (3.7)	Semantic context classification, multi-stakeholder trade-off modeling, and explicit epistemic uncertainty modeling ($\Delta U(A,C) \rightarrow \infty$).
MEASURE (Meas 1.1–4.3)	Quantitative/qualitative analysis, continuous performance tracking, goal-drift evaluation, and verification.	Sphere I, II, III Asynchronous Shadow Auditing AWI Sentry (3.8) AGI/ASI Verify (1.10/2.11)	KL-Divergence drift tracking ($\theta_{drift} = DKL(V_{target} V_{actual})$), zk-SNARK / zkVM polynomial proof circuits ($R = \{(x,w) \Phi(x,w)=1\}$).
MANAGE (Man 1.1–4.3)	Risk response execution, containment, escalation, emergency overrides, and incident recovery.	Hardware & CBP Circuit Breaker Protocol (CBP) Hardware NMI & Escrow AWI Void (3.7)	Hardware-level Non-Maskable Interrupts (NMI), 10ns/2-5ms Dual Kinetic Escrow Buffers, Triage Degradation Matrix, and Emergency Containment Protocol.

2. ISO/IEC 42001:2023 (AIMS) Control Clause Alignment

ISO/IEC 42001 establishes an actionable Management System for AI. TMGA aligns with the key normative clauses of ISO 42001 as detailed below:

ISO 42001 Clause	Standard Requirement	TMGA Architectural Feature	Compliance Verification
Clause 6.1 Actions to address AI risks & opportunities	Identify, assess, and document AI-specific risks with continuous risk calibration.	AWI Void (3.7) & Socratic Principle	Evaluates ethical confidence metric $\text{Conf}_{\text{ethical}}(A,C) < \tau_{\text{min}}$ to enforce operational pause or escalation when risk exceeds tolerance.
Clause 8.2 AI System Impact Assessment	Assess societal, ethical, and environmental impacts of consequential AI actions.	AWI Fold (3.5) & Context Committees	Multi-stakeholder trade-off optimization utilizing Integrated Axiology-MCDA framework to evaluate complex societal impacts.
Clause 8.4 Operational Control & Verification	Ensure system operates within defined policy constraints during production.	Cryptographic ZKP Pipeline	Generates Zero-Knowledge Certificates of Validity (zk-SNARKs) ensuring policy compliance without exposing internal model weights.
Clause 9.2 Internal Audit & Monitoring	Continuous performance evaluation and out-of-band monitoring without latency penalty.	Asynchronous Shadow Auditing	Snapshot isolation in ShadowRegistry with dynamic auditing interval ΔT shrinking exponentially during elevated risk states.
Clause 10.2 Nonconformity & Corrective Action	System containment, crisis arbitration, and forensic logging during failure events.	Circuit Breaker Protocol (CBP)	Quorum-activated crisis containment, append-only cryptographic hash chain logging, and hardware-level NMI actuator isolation.

3. Master 33-Dimension Regulatory Mapping Matrix

Every individual dimension within TMGA's three coexisting spheres (AGI Breadth, ASI Scale, AWI Meaning) corresponds to specific risk management, security, or audit controls within NIST AI RMF and ISO/IEC 42001.

Sphere & Dimension	Primary Cognitive Role	NIST RMF Mapping	ISO 42001 Mapping	Audit Artifact
1.1 AGI Mind	Cross-domain reasoning	MAP 1.1 / MANAGE 2.2	A.6.2 System Scope	Context Classification Trace
1.2 AGI Deep	Conceptual synthesis & abstraction	MAP 2.1 / MEASURE 2.1	A.8.2 Impact Assessment	Causal Abstraction Log
1.3 AGI Cortex	Executive functioning & memory	GOVERN 1.2 / MANAGE 1.1	A.6.1 Governance Structure	Executive Task Allocation
1.4 AGI Fold	Non-linear state space modeling	MAP 3.1 / MEASURE 2.2	A.8.3 Risk Treatment	High-Dim Trade-off Matrix
1.5 AGI Synapse	Modular signal routing	MANAGE 2.1 / MAP 1.2	A.7.2 Technical Resources	Agent Interconnect Trace
1.6 AGI Pulse	Persistent real-time state tracking	MEASURE 1.1 / MANAGE 1.3	A.9.1 Continuous Monitoring	State Telemetry Register
1.7 AGI Omni	Multimodal stream synthesis	MAP 1.3 / MEASURE 2.3	A.8.1 Data Quality & Prep	Multimodal Sanitization Log
1.8 AGI Guard	Sandboxed execution boundaries	MANAGE 1.2 / GOVERN 4.1	A.8.4 Boundary Control	Sandbox Isolation Proof
1.9 AGI Sentry	Behavioral auditing & drift detection	MEASURE 2.6 / MANAGE 3.1	A.9.2 Internal Audit	Behavioral Drift Alert
1.10 AGI Verify	Formal logic property verification	MEASURE 2.5 / GOVERN 3.2	A.8.5 Verification Test	Formal Logic Certificate
1.11 AGI Void	Epistemic uncertainty modeling	MAP 5.1 / MEASURE 1.2	A.6.1 Risk Identification	Uncertainty Boundary Flag
1.12 AGI Matter	Embodied physical interaction	MANAGE 4.1 / MAP 4.1	A.8.6 Physical Safety	Actuator Boundary Telemetry
1.13 AGI Wake	Recursive metacognitive assessment	GOVERN 2.1 / MEASURE 2.4	A.9.3 Management Review	Self-Assessment Diagnostic
2.1 ASI Mind	Hyper-scale cognitive throughput	MEASURE 2.1 / MAP 2.2	A.7.1 Compute Infrastructure	Hyper-Scale Execution Log
2.2 ASI Wake	Advanced scale state tracking	MEASURE 1.1 / MANAGE 1.3	A.9.1 Scale Monitoring	High-Throughput State Capture
2.3 ASI Deep	Automated hypothesis generation	MAP 2.3 / MEASURE 2.2	A.8.2 Data Synthesis	Scientific Discovery Audit
2.4 ASI Fold	Systemic complex optimization	MANAGE 2.3 / MAP 3.2	A.8.3 Systemic Risk Plan	Multi-Variable Opt Trace
2.5 ASI Thread	Distributed low-latency sync	MANAGE 2.1 / GOVERN 4.2	A.7.3 Network Resilience	Consensus Latency Log

Sphere & Dimension	Primary Cognitive Role	NIST RMF Mapping	ISO 42001 Mapping	Audit Artifact
2.6 ASI Bloom	Accelerated domain expansion	MAP 1.1 / MANAGE 1.1	A.6.2 Capabilities Growth	Frontier Exploration Boundary
2.7 ASI Dark	Opacity management & interpretability	MEASURE 2.7 / GOVERN 3.1	A.8.7 Transparency Control	zkVM Certificate Witness
2.8 ASI Guard	Cryptographic isolation barriers	MANAGE 1.2 / GOVERN 4.1	A.8.4 Sec Isolation	Cryptographic Barrier Lock
2.9 ASI Matter	High-precision physical interfaces	MANAGE 4.2 / MAP 4.2	A.8.6 Kinetic Safety	Hardware Interlock Status
2.10 ASI Void	Unmodeled systemic risk mitigation	MAP 5.2 / MANAGE 3.2	A.10.1 Incident Management	Systemic Fragility Sentinel
2.11 ASI Verify	Formal proof generation for scale	MEASURE 2.5 / GOVERN 3.2	A.8.5 Formal Proof	zk-SNARK Root Proof
3.1 AWI Mind	Context-aware semantic intent	MAP 1.1 / GOVERN 1.1	A.5.1 Alignment Policy	Semantic Intent Classification
3.2 AWI Core	Neural-symbolic hybrid integration	GOVERN 3.1 / MEASURE 2.5	A.8.5 Rule Consistency	Hybrid Constraint Evaluation
3.3 AWI Hub	Multi-agent & human-AI coord	GOVERN 2.2 / MANAGE 2.2	A.7.4 Stakeholder Coord	Human-in-the-Loop Audit
3.4 AWI Deep	Counterfactual outcome evaluation	MAP 2.2 / MEASURE 2.2	A.8.2 Counterfactual Test	Impact Simulation Report
3.5 AWI Fold	Societal & ecological trade-offs	GOVERN 1.2 / MAP 3.1	A.8.2 Societal Assessment	Integrated Axiology Matrix
3.6 AWI Safe	Constitutional guardrail enforcement	GOVERN 4.1 / MANAGE 1.2	A.6.1 Constitutional Guard	Safe-by-Design Rule Check
3.7 AWI Void	Epistemic pause & active restraint	MANAGE 3.1 / MAP 5.1	A.10.2 Operational Pause	Restraint Trigger & Confethical
3.8 AWI Sentry	Governance baseline drift tracking	MEASURE 2.6 / GOVERN 5.1	A.9.2 Policy Drift Audit	KL-Divergence Drift Report
3.9 AWI Systems	Institutional software translation	GOVERN 5.2 / MANAGE 1.1	A.5.2 Regulatory Alignment	Auditable Governance Log

4. Emergency Controls & Hardware-Assisted Enforcement

NIST AI RMF (MANAGE 3.2) and ISO 42001 (Clause 10.2) require hard containment mechanisms when software wrappers fail or when execution speeds exceed human cognitive bandwidth.

Control Mechanism	Activation Trigger	Technical Execution Architecture	Regulatory Assurance
Circuit Breaker Protocol (CBP)	Monotonic drift $\theta_{\text{drift}} > \tau_{\text{drift}}$ for 3 consecutive cycles AND $\partial\theta_{\text{drift}}/\partial t > 0$	Quorum-activated crisis layer assuming temporary coordination authority over 33 dimensions for max 72h window.	Prevents runaway goal-drift and systemic fragility under existential uncertainty.
Hardware-Assisted NMI & Escrow	Confethical(A,C) < τ_{min} OR software guardrail bypass detection	Processor-level Non-Maskable Interrupt (NMI) paired with 10ns instruction escrow & 2-5ms mechanical kinetic escrow buffers.	Eliminates execution leakage during post-NMI window; physically locks hardware actuators.
Bounded Operational Degradation	Computational stress or ZK-SNARK proving bottleneck approaching Δt_{max}	Pre-compiled, cryptographically sealed Triage Matrix: suspends heavy functions (ASI Bloom) while keeping core safety active (AGI Guard, AWI Void).	Prevents catastrophic system collapse by enforcing graceful, deterministic degradation.

5. Institutional Audit & Regulatory Implementation Roadmap

Auditor Guidance & Compliance Checklist:

- 1. Pre-Deployment (GOVERN & MAP):** Verify that AWI Systems (3.9) and Genesis Ethic rules are compiled into the CAWI boolean constraint function. Validate that task risk profiles generate deterministic ZK-SNARK risk proofs for validation committee routing.
- 2. Runtime Execution (MEASURE):** Inspect ShadowRegistry hash chains for append-only audit trail integrity. Ensure dynamic auditing interval ΔT correctly scales with system operational tempo.
- 3. Crisis Preparedness (MANAGE):** Conduct physical NMI interlock tests and verify that the 10ns / 2-5ms dual escrow buffers flush pending actuator commands upon intervention signal trigger.